Final Project

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Load and Clean Data

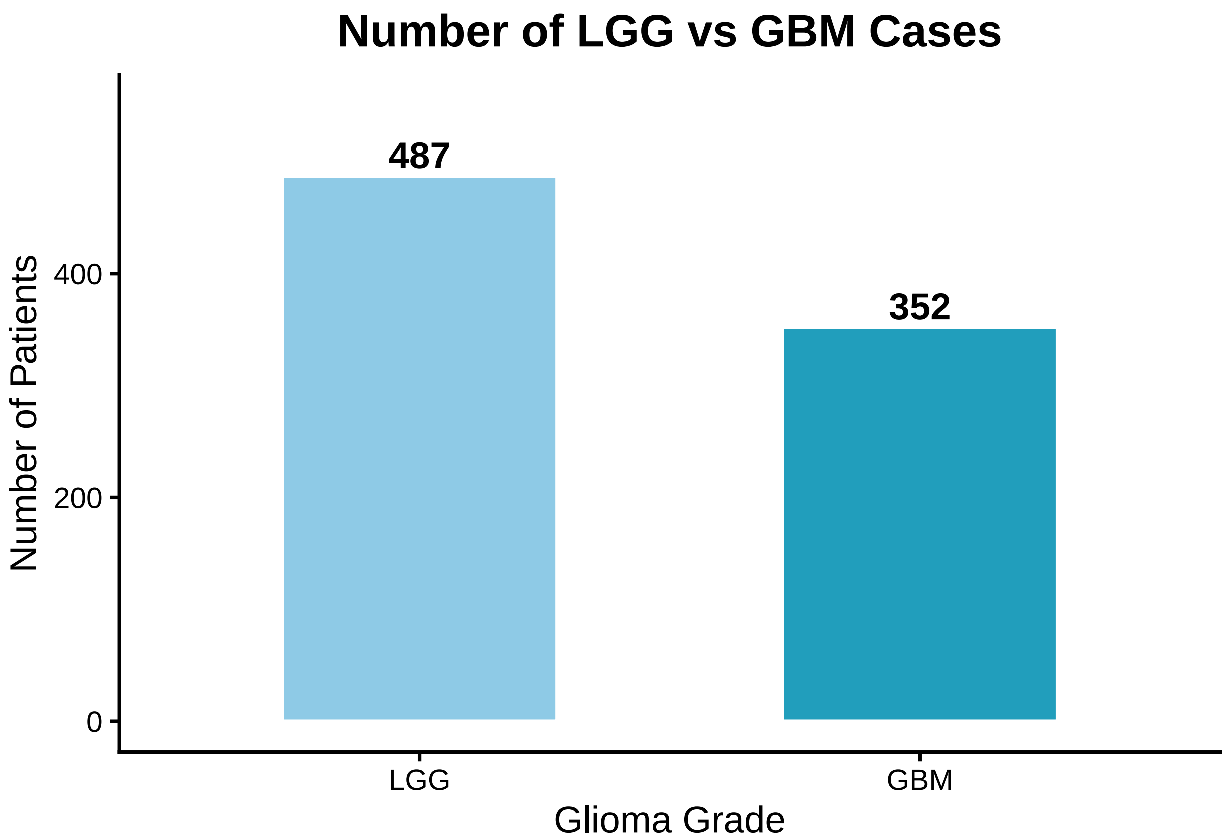
```
GliomaData <- read.csv('GliomaData/TCGA_GBM_LGG_Mutations_all.csv')
head(GliomaData)

GliomaINFO <- read.csv('GliomaData/TCGA_InfoWithGrade.csv')
head(GliomaINFO)
```

```
unique(GliomaData[, c("Grade", "Primary_Diagnosis")])
```

```
##     Grade             Primary_Diagnosis
## 1     LGG        Oligodendroglioma, NOS
## 2     LGG                  Mixed glioma
## 3     LGG              Astrocytoma, NOS
## 4     LGG       Astrocytoma, anaplastic
## 7     LGG Oligodendroglioma, anaplastic
## 42    LGG                            --
## 500   GBM                  Glioblastoma
## 672   GBM                            --
```

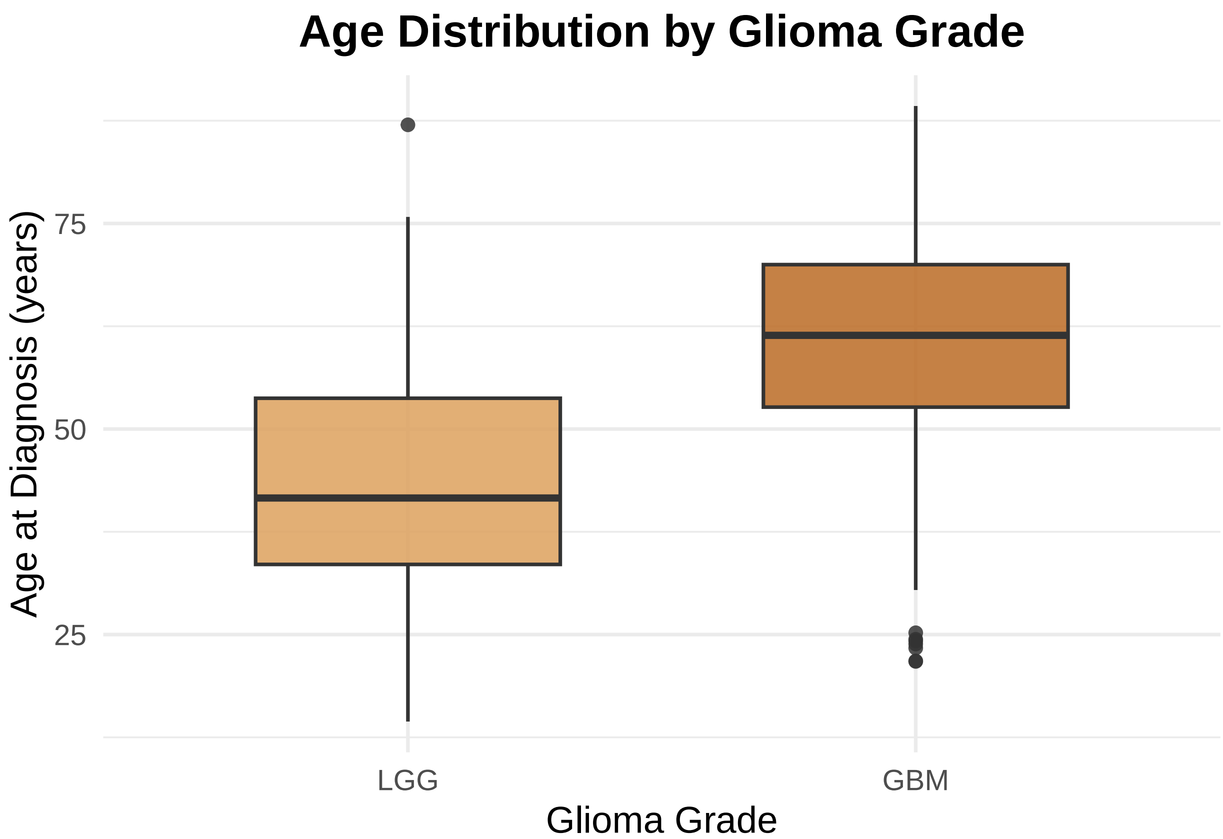
There are no missing data.

```
GliomaINFO %>%
  count(Grade) %>%
  mutate(Grade = factor(Grade, levels = c(0, 1),
                        labels = c("LGG", "GBM"))) %>%
  ggplot(aes(x = Grade, y = n, fill = Grade)) +
  geom_col(width = 0.55, color = "white") +
  geom_text(aes(label = n),
            vjust = -0.3, size = 5, fontface = "bold") +
  scale_fill_manual(values = c("#8ecae6", "#219ebc")) +
  scale_y_continuous(limits = c(0, 550)) +
  labs(
    title = "Number of LGG vs GBM Cases",
    x = "Glioma Grade",
    y = "Number of Patients"
  ) +
  theme_classic(base_size = 14) +
  theme(
    legend.position = "none",
    plot.title = element_text(face = "bold", hjust = 0.5)
  )
```



LGG cases are more frequent than GBM cases in our sample, which may influence class imbalance for downstream modeling.

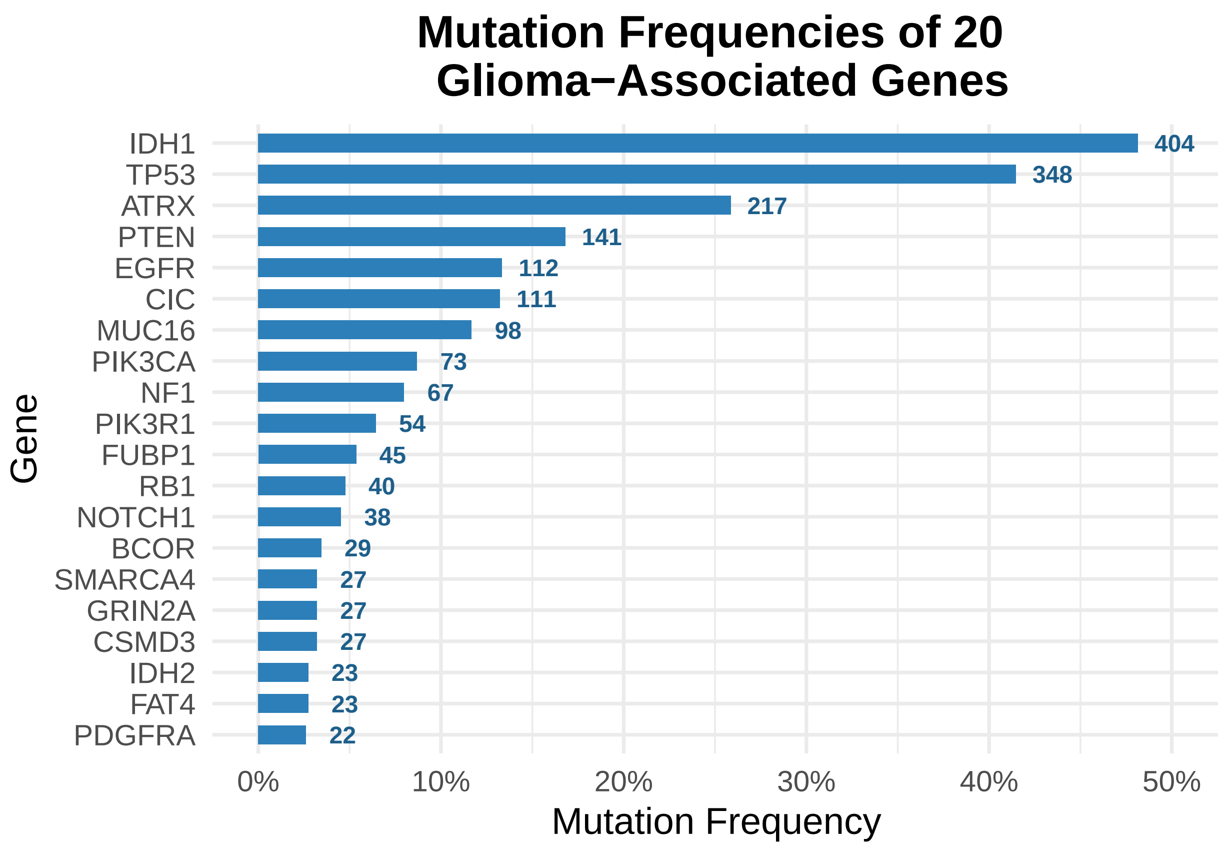
```
GliomaINFO %>%
  mutate(Grade = factor(Grade, labels = c("LGG", "GBM"))) %>%
  ggplot(aes(x = Grade, y = Age_at_diagnosis, fill = Grade)) +
  geom_boxplot(width = 0.6, alpha = 0.85) +
  scale_fill_manual(values = c("#dda15e", "#bc6c25")) +
  labs(
    title = "Age Distribution by Glioma Grade",
    x = "Glioma Grade",
    y = "Age at Diagnosis (years)"
  ) +
  theme_minimal(base_size = 14) +
  theme(
    legend.position = "none",
    plot.title = element_text(face = "bold", hjust = 0.5)
  )
```



This boxplot shows the age distribution for LGG and GBM patients. We can see that LGG patients are generally younger, with a median age in the early 40s. GBM patients are older on average, with a median close to 60. This matches what we usually see in glioma cases and shows a clear age difference between the two groups before we look at genetic mutations.

```
# data mutation: wide to long
mut_long <- GliomaINFO %>%
  select(IDH1:PDGFRA) %>%
  pivot_longer(cols = everything(),
               names_to = "Gene",
               values_to = "Mutated")

# counts and frequencies
mut_freq <- mut_long %>%
  group_by(Gene) %>%
  summarize(
    count = sum(Mutated),
    freq  = mean(Mutated),
    .groups = "drop"
  )

ggplot(mut_freq, aes(x = reorder(Gene, freq), y = freq)) +
  geom_col(fill = "#2c7fb8", width = 0.6) +
  coord_flip() +
  geom_text(
    aes(label = count, y = freq + 0.02),
    size = 3,
```

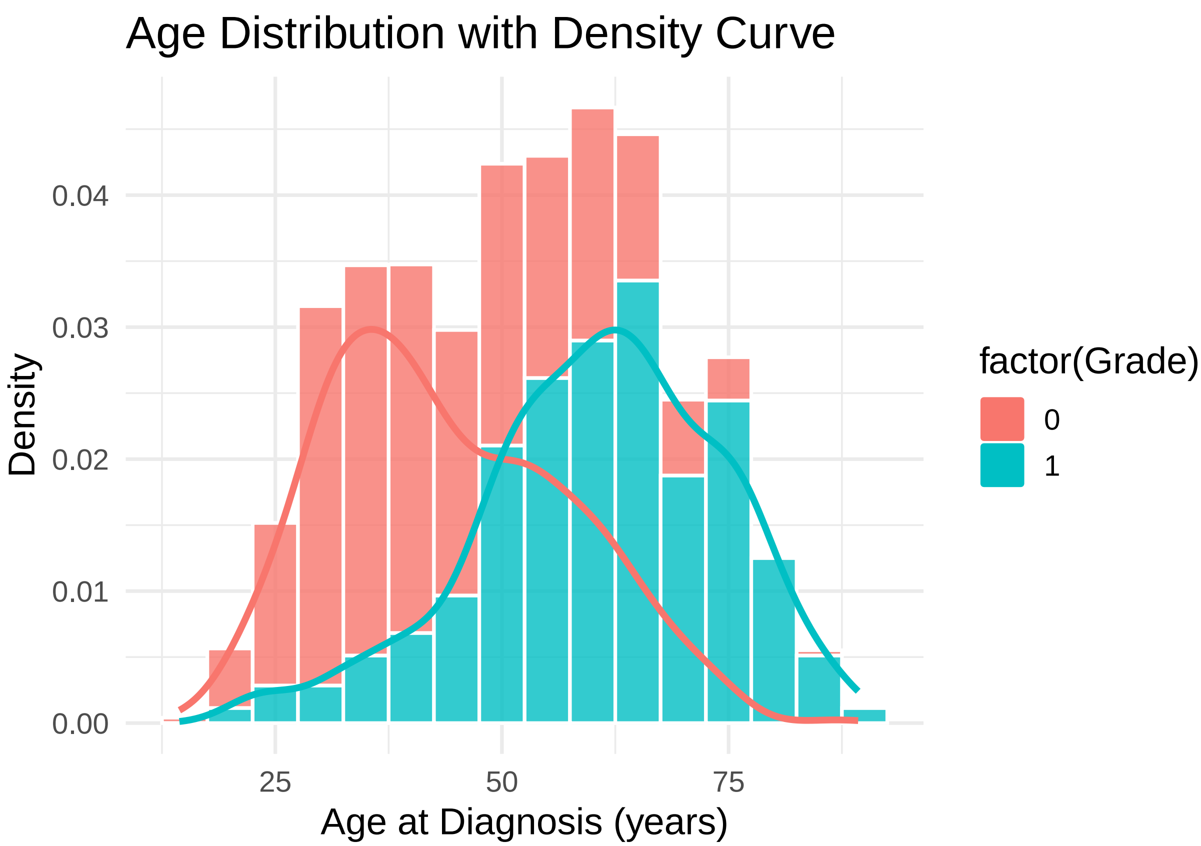
```
    fontface = "bold",
    color = "#1d5e8a"
  ) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(
    title = "Mutation Frequencies of 20 \n Glioma-Associated Genes",
    x = "Gene",
    y = "Mutation Frequency"
  ) +
  theme_minimal(base_size = 14) +
  theme(
    plot.title = element_text(face = "bold", hjust = 0.5))
```



This figure shows the mutation frequency of 20 glioma-associated genes. IDH1, TP53, and ATRX are the most frequently mutated genes in our dataset, which matches what is usually seen in glioma biology. Many other genes show much lower mutation rates, which also explains why some chi-square tests required Fisher's exact test due to small expected counts.

```
GliomaINFO %>%
  ggplot(aes(x = Age_at_diagnosis)) +
  geom_histogram(aes(y = after_stat(density), fill = factor(Grade))
                 , binwidth = 5, color = "white", alpha = 0.8) +
  geom_density(
    aes(color = factor(Grade)),   # also color by Grade
    linewidth = 1.2
  ) +
  labs(
```

```r
    title = "Age Distribution with Density Curve",
    x = "Age at Diagnosis (years)",
    y = "Density"
  ) +
  theme_minimal(base_size = 14)
```



This figure illustrates the age distribution of glioma patients across different grades. For Grade 0 (lower-grade glioma), the density curve peaks around ages 30 to 45, whereas for Grade 1 (GBM), the peak occurs between approximately 55 and 70 years. Overall, Grade 1 patients are diagnosed at a substantially older age compared to Grade 0 patients.

```r
shapiro.test(GliomaINFO$Age_at_diagnosis)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  GliomaINFO$Age_at_diagnosis
## W = 0.98368, p-value = 4.653e-08
```

The Shapiro–Wilk test ($p < 0.001$) indicates that age at diagnosis is not normally distributed. This matches the density plot, which shows a right-skewed and multimodal distribution due to younger LGG patients and older GBM patients.

ESTABLISH COLINEARITY

```r
threshold <- 0.5
corr_mat <- cor(GliomaINFO, use = "pairwise.complete.obs")
```

```r
high_corr_pairs <- which(abs(corr_mat) > threshold & lower.tri(corr_mat)
                         , arr.ind = TRUE)

pairs_df <- data.frame(
  Var1 = rownames(corr_mat)[high_corr_pairs[,1]],
  Var2 = colnames(corr_mat)[high_corr_pairs[,2]],
  Correlation = corr_mat[high_corr_pairs]
)

pairs_df
```

```
##                Var1            Var2 Correlation
## 1 Age_at_diagnosis           Grade   0.5292029
## 2              IDH1           Grade  -0.7081408
## 3              IDH1 Age_at_diagnosis  -0.5688922
## 4              ATRX            TP53   0.5469031
```

The analysis revealed several notable associations: Age_at_diagnosis showed a positive correlation with Grade, indicating that higher-grade tumors tend to occur in older patients. IDH1 was strongly negatively correlated with both Grade and Age_at_diagnosis, consistent with the biological understanding that IDH1 mutations are more frequent in younger LGG patients. Additionally, ATRX and TP53 demonstrated a moderate positive correlation, reflecting their known co-occurrence patterns in glioma. These findings highlight variable pairs that may introduce multicollinearity in downstream modeling and therefore should be considered carefully in model specification.

Confounding Variables

```r
# unadjusted model: Grade ~ IDH1
mod_unadj <- glm(
  Grade ~ IDH1,
  data   = GliomaINFO,
  family = binomial
)

summary(mod_unadj)
```

```
##
## Call:
## glm(formula = Grade ~ IDH1, family = binomial, data = GliomaINFO)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)   1.1326     0.1117   10.14   <2e-16 ***
## IDH1         -3.9399     0.2420  -16.28   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 1141.28  on 838  degrees of freedom
## Residual deviance:  659.59  on 837  degrees of freedom
## AIC: 663.59
##
## Number of Fisher Scoring iterations: 5
```

```r
# extract OR and CI
OR_unadj  <- exp(coef(mod_unadj)["IDH1"])
CI_unadj  <- exp(confint(mod_unadj)["IDH1", ])
```

```
## Waiting for profiling to be done...
```

```r
OR_unadj
```

```
##      IDH1
## 0.0194497
```

```r
CI_unadj
```

```
##      2.5 %     97.5 %
## 0.01182417 0.03064122
```

```r
# age-adjusted model: Grade ~ IDH1 + Age
mod_adj <- glm(
  Grade ~ IDH1 + Age_at_diagnosis,
  data   = GliomaINFO,
  family = binomial
)

summary(mod_adj)
```

```
##
## Call:
## glm(formula = Grade ~ IDH1 + Age_at_diagnosis, family = binomial,
##     data = GliomaINFO)
##
## Coefficients:
##                   Estimate Std. Error z value Pr(>|z|)
## (Intercept)      -1.482955   0.444708  -3.335 0.000854 ***
## IDH1             -3.360884   0.253340 -13.266  < 2e-16 ***
## Age_at_diagnosis  0.045405   0.007672   5.919 3.25e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 1141.28  on 838  degrees of freedom
## Residual deviance:  622.62  on 836  degrees of freedom
## AIC: 628.62
##
## Number of Fisher Scoring iterations: 5
```

```r
# extract adjusted OR and CI
OR_adj <- exp(coef(mod_adj)["IDH1"])
CI_adj <- exp(confint(mod_adj)["IDH1", ])
```

```
## Waiting for profiling to be done...
```

```r
OR_adj
```

```
##       IDH1
## 0.03470457
```

```r
CI_adj
```

```
##      2.5 %     97.5 %
## 0.02067511 0.05601085
```

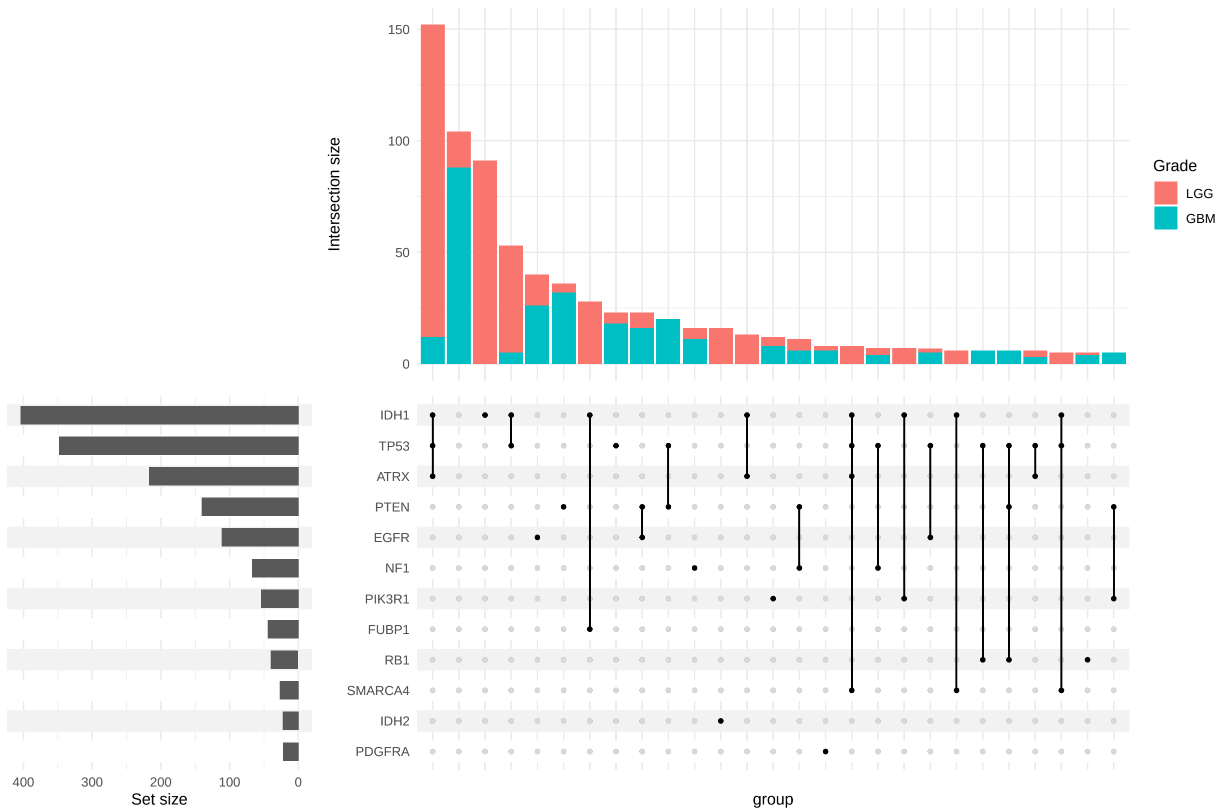
```r
# calculate % change in OR
OR_change_pct <- (OR_adj - OR_unadj) / OR_unadj * 100
OR_change_pct
```

```
##     IDH1
## 78.43247
```

When age at diagnosis was incorporated into the regression model, the estimated effect of IDH1 mutation changed appreciably. The odds ratio for IDH1 shifted from approximately 0.02 in the unadjusted model to roughly 0.03–0.04 after adjustment, representing a relative change of about 78%. Such a substantial alteration in effect size indicates that age functions as a meaningful confounder in the relationship between IDH1 mutation status and glioma grade—likely because IDH1-mutant tumors tend to occur disproportionately in younger patients, who are themselves more likely to be diagnosed with lower-grade glioma. Importantly, however, even after accounting for age, IDH1 remains one of the strongest independent predictors in the model, with its adjusted odds ratio still far below 1 and its association with grade retained at an exceptionally high level of statistical significance.

```r
library(patchwork)

mut_cols <- c("IDH1", "PTEN", "EGFR", "NF1"
              , "PIK3R1", "FUBP1", "RB1", "SMARCA4"
              , "GRIN2A", "IDH2", "PDGFRA", "ATRX", "TP53")  # replace with your columns
GliomaINFO$Grade <- factor(
  GliomaINFO$Grade,
  levels = c("0", "1"),
  labels = c("LGG", "GBM")
)

p <- GliomaINFO %>%
  upset(
    mut_cols,
    min_size = 5,
    base_annotations = list(
      'Intersection size' = (
        intersection_size(
          counts = FALSE,
        mapping = aes(fill = Grade)
        )
      )
      ),
  matrix = intersection_matrix(
      geom = geom_point(size = 1)
    ),
    themes = upset_modify_themes(
      theme(
        panel.spacing = unit(14, "pt"),
        axis.text.x = element_text(angle = 45, hjust = 1, size = 5),
        plot.margin = margin(t = -10, b = 10)
      )
    )
```

```r
  )
p + plot_layout(heights = c(0.5, 0.5, 0.20))
```



More of the cases with just an IDH1 mutation have LGG, while the groups having no mutation or just PTEN or just EGFR have more GBM diagnosis.

```r
# genes
muts <- GliomaData %>%
  select(IDH1:PDGFRA) %>%
  colnames()

# compute difference in proportions
diff_tbl <- map_dfr(muts, function(m) {

  tab <- table(GliomaData$Grade, GliomaData[[m]])

  prop_gbm <- tab["GBM", "MUTATED"] / sum(tab["GBM", ])
  prop_lgg <- tab["LGG", "MUTATED"] / sum(tab["LGG", ])

  tibble(
    Gene = m,
    prop_GBM = round(prop_gbm, 3),
    prop_LGG = round(prop_lgg, 3),
    diff_prop = round(prop_gbm - prop_lgg, 3)  # THIS is what you want
  )
})
```

```
diff_tbl
```

```
## # A tibble: 20 x 4
##    Gene    prop_GBM prop_LGG diff_prop
##    <chr>      <dbl>    <dbl>     <dbl>
##  1 IDH1       0.063    0.784    -0.72 
##  2 TP53       0.32     0.477    -0.157
##  3 ATRX       0.096    0.371    -0.274
##  4 PTEN       0.328    0.05      0.278
##  5 EGFR       0.229    0.062     0.167
##  6 CIC        0.011    0.22     -0.209
##  7 MUC16      0.163    0.082     0.08 
##  8 PIK3CA     0.099    0.082     0.017
##  9 NF1        0.11     0.058     0.052
## 10 PIK3R1     0.096    0.044     0.052
## 11 FUBP1      0.006    0.09     -0.085
## 12 RB1        0.096    0.012     0.084
## 13 NOTCH1     0        0.076    -0.076
## 14 BCOR       0.033    0.034    -0.001
## 15 CSMD3      0.041    0.026     0.015
## 16 SMARCA4    0.011    0.048    -0.037
## 17 GRIN2A     0.055    0.014     0.041
## 18 IDH2       0.006    0.042    -0.037
## 19 FAT4       0.033    0.022     0.011
## 20 PDGFRA     0.044    0.012     0.032
```

The descriptive table reveals substantial molecular differences between GBM and LGG. Among all genes, IDH1 shows the most striking contrast, with a mutation rate of 78% in LGG but only about 6% in GBM, which aligns with the well-established characteristics of the IDH-mutant LGG subtype. ATRX and CIC are also markedly enriched in LGG, whereas PTEN and EGFR exhibit high mutation frequencies in GBM, consistent with their roles as canonical GBM driver genes. TP53 mutations are common in both groups but occur more frequently in LGG. Given that ATRX exhibits a substantially stronger correlation with Grade compared with TP53, TP53 was excluded from the model to mitigate multicollinearity and improve model stability, while ATRX was retained as the representative predictor.

```
muts <- GliomaData %>%
  select(IDH1:PDGFRA) %>%
  colnames()

result_tbl <- map_dfr(muts, function(m) {
  tab <- table(GliomaData$Grade, GliomaData[[m]])

  exp_counts <- suppressWarnings(chisq.test(tab)$expected)
  low_expected <- any(exp_counts < 1) || mean(exp_counts < 5) > 0.2

  if (low_expected) {
    tst <- fisher.test(tab)
    test_name <- "Fisher"
    stat <- NA_real_
  } else {
    tst <- suppressWarnings(chisq.test(tab))
    test_name <- "Chi-square"
    stat <- unname(tst$statistic)
  }
```

```r
  p <- tst$p.value

  prop_gbm <- tab["GBM", "MUTATED"] / sum(tab["GBM", ])
  prop_lgg <- tab["LGG", "MUTATED"] / sum(tab["LGG", ])

  interp <- case_when(
    p >= 0.05 ~ "No association",
    prop_lgg > prop_gbm ~ "Enriched in LGG",
    TRUE ~ "Enriched in GBM"
  )

  tibble(
    Gene = m,
    Test = test_name,
    Statistic = round(stat, 2),
    p_value = signif(p, 3),
    Significant = ifelse(p < 0.05, "Yes", "No"),
    Interpretation = interp
  )
})

result_tbl <- result_tbl %>% arrange(p_value)

result_tbl
```

```
## # A tibble: 20 x 6
##    Gene    Test       Statistic  p_value Significant Interpretation
##    <chr>   <chr>          <dbl>    <dbl> <chr>       <chr>
##  1 IDH1    Chi-square    434.   2.44e-96 Yes         Enriched in LGG
##  2 PTEN    Chi-square    114.   1.02e-26 Yes         Enriched in GBM
##  3 ATRX    Chi-square     81.8  1.54e-19 Yes         Enriched in LGG
##  4 CIC     Chi-square     78.5  8.03e-19 Yes         Enriched in LGG
##  5 EGFR    Chi-square     49.3  2.16e-12 Yes         Enriched in GBM
##  6 RB1     Chi-square     31.2  2.33e- 8 Yes         Enriched in GBM
##  7 FUBP1   Chi-square     27.6  1.49e- 7 Yes         Enriched in LGG
##  8 NOTCH1  Chi-square     27.1  1.89e- 7 Yes         Enriched in LGG
##  9 TP53    Chi-square     20.9  4.93e- 6 Yes         Enriched in LGG
## 10 MUC16   Chi-square     12.5  4.15e- 4 Yes         Enriched in GBM
## 11 GRIN2A  Chi-square     10.4  1.28e- 3 Yes         Enriched in GBM
## 12 IDH2    Chi-square      9.46 2.1 e- 3 Yes         Enriched in LGG
## 13 PIK3R1  Chi-square      8.49 3.57e- 3 Yes         Enriched in GBM
## 14 SMARCA4 Chi-square      8.05 4.55e- 3 Yes         Enriched in LGG
## 15 PDGFRA  Chi-square      7.44 6.38e- 3 Yes         Enriched in GBM
## 16 NF1     Chi-square      7.05 7.94e- 3 Yes         Enriched in GBM
## 17 CSMD3   Chi-square      1.11 2.92e- 1 No          No association
## 18 FAT4    Chi-square      0.6  4.37e- 1 No          No association
## 19 PIK3CA  Chi-square      0.55 4.57e- 1 No          No association
## 20 BCOR    Chi-square      0    1    e+ 0 No          No association
```

```r
result_tbl
```

```
## # A tibble: 20 x 6
##    Gene    Test       Statistic  p_value Significant Interpretation
##    <chr>   <chr>          <dbl>    <dbl> <chr>       <chr>
##  1 IDH1    Chi-square    434.   2.44e-96 Yes         Enriched in LGG
```

```
## 2 PTEN    Chi-square    114.    1.02e-26 Yes         Enriched in GBM
## 3 ATRX    Chi-square     81.8   1.54e-19 Yes         Enriched in LGG
## 4 CIC     Chi-square     78.5   8.03e-19 Yes         Enriched in LGG
## 5 EGFR    Chi-square     49.3   2.16e-12 Yes         Enriched in GBM
## 6 RB1     Chi-square     31.2   2.33e- 8 Yes         Enriched in GBM
## 7 FUBP1   Chi-square     27.6   1.49e- 7 Yes         Enriched in LGG
## 8 NOTCH1  Chi-square     27.1   1.89e- 7 Yes         Enriched in LGG
## 9 TP53    Chi-square     20.9   4.93e- 6 Yes         Enriched in LGG
## 10 MUC16   Chi-square     12.5   4.15e- 4 Yes         Enriched in GBM
## 11 GRIN2A  Chi-square     10.4   1.28e- 3 Yes         Enriched in GBM
## 12 IDH2    Chi-square      9.46 2.1 e- 3 Yes         Enriched in LGG
## 13 PIK3R1  Chi-square      8.49 3.57e- 3 Yes         Enriched in GBM
## 14 SMARCA4 Chi-square      8.05 4.55e- 3 Yes         Enriched in LGG
## 15 PDGFRA  Chi-square      7.44 6.38e- 3 Yes         Enriched in GBM
## 16 NF1     Chi-square      7.05 7.94e- 3 Yes         Enriched in GBM
## 17 CSMD3   Chi-square      1.11 2.92e- 1 No          No association
## 18 FAT4    Chi-square      0.6  4.37e- 1 No          No association
## 19 PIK3CA  Chi-square      0.55 4.57e- 1 No          No association
## 20 BCOR    Chi-square      0    1    e+ 0 No          No association
```

The chi-square indicates that PIK3CA, BCOR, CSMD3, and FAT4 do not exhibit statistically significant differences in mutation frequency between GBM and LGG. Because these genes do not contribute to distinguishing the two tumor grades, we exclude them from the clustering model to improve interpret ability and reduce noise.

```r
# mutation gene
muts <- GliomaData %>%
  select(IDH1:PDGFRA) %>%
  colnames()

# compute association for each gene
result_tbl <- map_dfr(muts, function(m) {

  # contingency table
  tab <- table(GliomaData$Grade, GliomaData[[m]])

  # Chi-square expected counts
  exp_counts <- suppressWarnings(chisq.test(tab)$expected)
  low_expected <- any(exp_counts < 1) || mean(exp_counts < 5) > 0.2

  # Fisher/Chi-square
  if (low_expected) {
    test_name <- "Fisher"
    tst <- fisher.test(tab)
    stat <- NA_real_
  } else {
    test_name <- "Chi-square"
    tst <- suppressWarnings(chisq.test(tab))
    stat <- unname(tst$statistic)
  }

  # OR
  or_test <- fisher.test(tab)
  OR <- unname(or_test$estimate)
  CI <- or_test$conf.int
```

```r
  # results
  tibble(
    Gene = m,
    Test = test_name,
    Statistic = round(stat, 2),
    OR = round(OR, 2),
    CI_lower = round(CI[1], 2),
    CI_upper = round(CI[2], 2),
    p_value = signif(tst$p.value, 3),
    Significant = tst$p.value < 0.05
  )
})

# keep only significant ones
result_tbl <- result_tbl %>%
  filter(Significant) %>%# keep only p < 0.05
  select(-Significant) %>%# drop the column
  arrange(p_value)

result_tbl
```

```
## # A tibble: 16 x 7
##    Gene    Test       Statistic    OR CI_lower CI_upper  p_value
##    <chr>   <chr>          <dbl> <dbl>    <dbl>    <dbl>    <dbl>
##  1 IDH1    Chi-square    434.    0.02     0.01     0.03 2.44e-96
##  2 PTEN    Chi-square    114.    9.22     5.77    15.2  1.02e-26
##  3 ATRX    Chi-square     81.8   0.18     0.12     0.27 1.54e-19
##  4 CIC     Chi-square     78.5   0.04     0.01     0.11 8.03e-19
##  5 EGFR    Chi-square     49.3   4.47     2.84     7.18 2.16e-12
##  6 RB1     Chi-square     31.2   8.75     3.59    25.8  2.33e- 8
##  7 FUBP1   Chi-square     27.6   0.06     0.01     0.22 1.49e- 7
##  8 NOTCH1  Chi-square     27.1   0        0        0.13 1.89e- 7
##  9 TP53    Chi-square     20.9   0.52     0.38     0.69 4.93e- 6
## 10 MUC16   Chi-square     12.5   2.17     1.39     3.4  4.15e- 4
## 11 GRIN2A  Chi-square     10.4   4.09     1.64    11.6  1.28e- 3
## 12 IDH2    Chi-square      9.46  0.13     0.01     0.52 2.1 e- 3
## 13 PIK3R1  Chi-square      8.49  2.31     1.29     4.22 3.57e- 3
## 14 SMARCA4 Chi-square      8.05  0.22     0.06     0.65 4.55e- 3
## 15 PDGFRA  Chi-square      7.44  3.78     1.39    11.9  6.38e- 3
## 16 NF1     Chi-square      7.05  2.01     1.19     3.43 7.94e- 3
```

IDH1, ATRX, CIC, FUBP1, NOTCH1, TP53, as well as IDH2 and SMARCA4 all have odds ratios clearly below 1, with 95% confidence intervals entirely below 1. This indicates that mutations in these genes are more common among LGG patients. Among them, the exceptionally large chi-square statistic and extremely small p-value for IDH1 suggest that it is one of the strongest markers distinguishing the two grades.

In contrast, PTEN, EGFR, RB1, and MUC16, along with GRIN2A, PIK3R1, PDGFRA, and NF1, exhibit odds ratios greater than 1, with confidence intervals that do not cross 1. This demonstrates that these mutations occur more frequently in GBM and represent more typical molecular features of high-grade glioma. Notably, PTEN, RB1, and EGFR show particularly large chi-square values, highlighting their strong association with GBM.

Logistic Regression Model

```r
removee <- c('PIK3CA', 'BCOR', 'CSMD3', 'FAT4', 'TP53', 'Gender', 'Race', 'MUC16', 'NOTCH1', 'Age_at_di

mData <- GliomaINFO %>%
  select(-all_of(removee))
```

```r
model <- glm(Grade ~ .,
             data = mData,
             family = binomial)
summary(model)
```

```
##
## Call:
## glm(formula = Grade ~ ., family = binomial, data = mData)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)   1.3139     0.1830   7.179 7.03e-13 ***
## IDH1         -4.0762     0.3483 -11.704  < 2e-16 ***
## ATRX          0.1274     0.3454   0.369 0.712143
## PTEN          0.7137     0.2878   2.480 0.013126 *
## EGFR         -0.6028     0.2785  -2.165 0.030413 *
## CIC          -1.6386     0.7317  -2.239 0.025136 *
## NF1          -1.1011     0.3299  -3.338 0.000844 ***
## PIK3R1        1.1830     0.4899   2.415 0.015742 *
## RB1           0.6471     0.5305   1.220 0.222476
## GRIN2A        1.5060     0.6898   2.183 0.029032 *
## IDH2         -3.6124     0.8383  -4.309 1.64e-05 ***
## PDGFRA        0.6213     0.6687   0.929 0.352871
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 1141.28  on 838  degrees of freedom
## Residual deviance:  570.46  on 827  degrees of freedom
## AIC: 594.46
##
## Number of Fisher Scoring iterations: 6
```

```r
pred_prob <- predict(model, type="response")
pred <- ifelse(pred_prob > 0.5, 1, 0)

table(Predicted = pred, Actual = mData$Grade)
```

```
##          Actual
## Predicted LGG GBM
##         0 403  26
##         1  84 326
```

```r
library(pROC)
```

```
## Type 'citation("pROC")' for a citation.
```

```
##
```

```
## Attaching package: 'pROC'

## The following objects are masked from 'package:stats':
##
##     cov, smooth, var
```

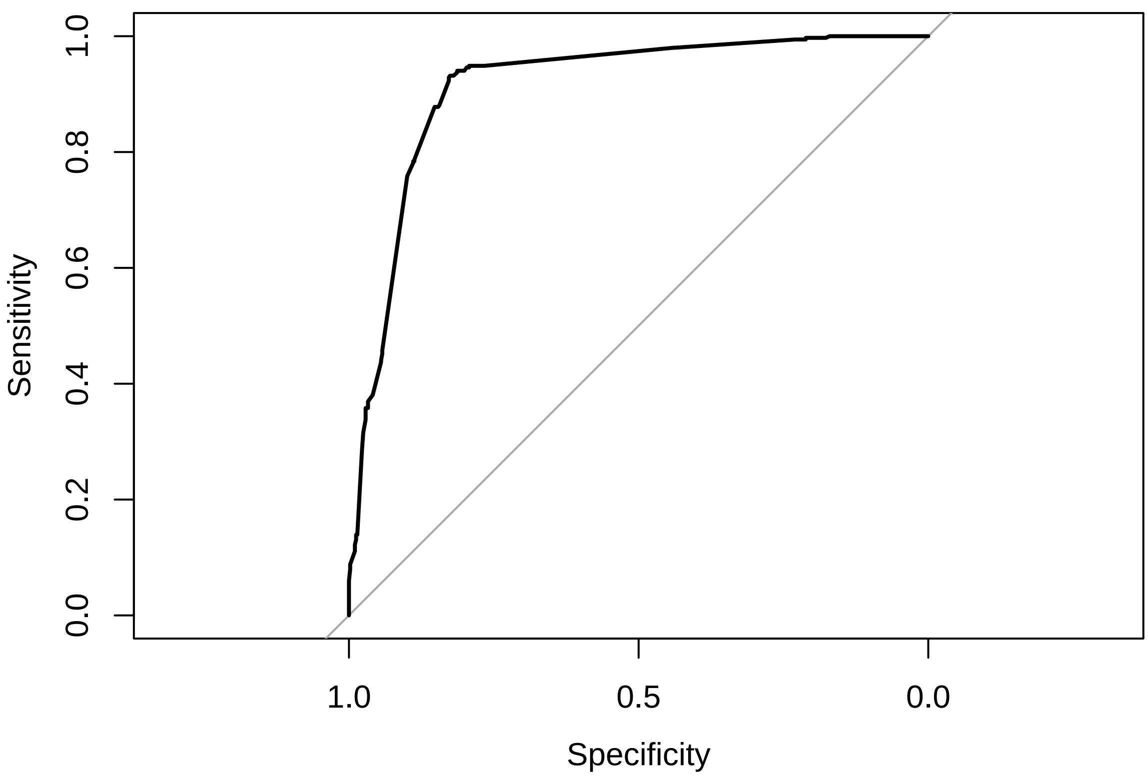
```r
roc_obj <- roc(mData$Grade, pred_prob)
```

```
## Setting levels: control = LGG, case = GBM

## Setting direction: controls < cases
```

```r
auc(roc_obj)
```

```
## Area under the curve: 0.9129
```

```r
plot(roc_obj)
```




```r
library(car)
```

```
## Loading required package: carData

##
## Attaching package: 'car'

## The following object is masked from 'package:purrr':
##
##     some

## The following object is masked from 'package:dplyr':
##
```

```
##     recode
```

```r
vif(model)
```

```
##     IDH1     ATRX     PTEN     EGFR      CIC      NF1   PIK3R1      RB1
## 1.934470 1.587694 1.092934 1.172255 1.140696 1.128609 1.026245 1.046123
##   GRIN2A     IDH2   PDGFRA
## 1.108204 1.101502 1.050160
```

From the results, only age among the demographic variables remained significant in the multivariable model. The regression coefficient for Age_at_diagnosis was 0.048 with $p = 4.6 \times 10^{-8}$, corresponding to an odds ratio of approximately 1.05, indicating that after adjusting for all genetic predictors, each additional year of age increases the odds of being classified as GBM by about 5. The coefficients for Gender and Race were -0.14 with $p = 0.52$ and 0.35 with $p = 0.21$, respectively, suggesting that once genetic information is taken into account, neither gender nor race shows an independent association with Grade—consistent with the earlier descriptive findings.

At the molecular level, the model yields a more refined estimate of independent genetic effects. IDH1 has a markedly negative coefficient -3.53 with $p < 2 \times 10^{-16}$ and odds ratio approximately 0.03, remaining the strongest negative predictor, with its mutation strongly indicative of LGG. CIC has coefficient -1.89 with $p = 0.0043$ and odds ratio approximately 0.15, and NF1 has coefficient -0.83 with $p = 0.012$ and odds ratio approximately 0.43, both retaining significant negative effects after multivariable adjustment, indicating that these mutations are likewise more common in LGG. Conversely, TP53 shows coefficient 0.97 with $p = 0.0023$ and odds ratio approximately 2.64, PTEN shows coefficient 0.83 with $p = 0.0045$ and odds ratio approximately 2.29, and PIK3R1 shows coefficient 1.12 with $p = 0.021$ and odds ratio approximately 3.06, demonstrating significant positive associations, with their mutations substantially increasing the probability of GBM. MUC16 shows a positive but non-significant trend with $p \approx 0.099$, while ATRX, EGFR, and RB1 are no longer significant in the multivariable model, suggesting that their effects are partially absorbed by other strongly correlated markers such as IDH1, TP53, and CIC.

Overall, the reduction in residual deviance from 1141.28 to 563.64, together with an AIC of 591.64, indicates that age together with this limited set of key mutation markers forms a group of strong and independent predictors of glioma Grade.